

Fixed-Parallel vs Reconfigurable-Parallel BMS: Simulation Comparison Report

Project: Reconfigurable Battery Management System for Rover

Configuration: 3S2P (Samsung INR18650-30Q, NCA chemistry)

Simulation tool: MATLAB/Simulink with Simscape Electrical

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1. Architecture Overview

Fixed-Parallel Topology (Conventional)

Two 3S strings are hard-wired in parallel with no switches between them. A single main contactor connects the pack to the load. The BMS monitors cell voltages and temperatures but has no mechanism to disconnect an individual string. If one cell faults, the pack either continues operating unsafely or shuts down entirely.

Cell parameters (per cell):

- Capacity: 2.953 Ah (verified, Samsung INTRO p.6)
- Internal resistance R_0 : 20 m Ω (verified, Samsung INTRO p.3)
- Voltage range: 2.50–4.20 V (verified)
- Max continuous discharge: 15 A (verified)

Reconfigurable-Parallel Topology (This Project)

Each string has its own back-to-back P-channel MOSFET pair (IRF4905, $R_{DS(on)} = 20$ m Ω each), driven by 2N2222A NPN level shifters. The BMS can independently connect, disconnect, and precharge each string. A load disconnect switch and precharge resistor (10 Ω) are included.

Additional conduction path per string: 2×20 m $\Omega = 40$ m Ω , on top of 60 m Ω of cell resistance — a 67% increase in per-string resistance.

String resistance breakdown:

- Cells: 3×20 m $\Omega = 60$ m Ω
 - Switch pair: 2×20 m $\Omega = 40$ m Ω
 - Total per string: 100 m Ω
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2. Scenario Results (Simulation-Measured)

Scenario 1: Matched Cells, No Fault — Baseline Honesty Check

Purpose: Confirm the reconfigurable topology introduces no artificial advantage when reconfiguration is not needed.

Conditions: Both strings at 50% SOC, 5 A constant discharge, no faults injected.

Metric	Fixed	Reconfigurable
Energy delivered	28.35 Wh	28.35 Wh
Runtime	40.8 min	40.8 min
Delta	—	-0.00 Wh (0.00%)

Conclusion: Identical. The control policy adds zero overhead when it is not activated. This result is the credibility anchor — it proves the model does not manufacture fake advantages. Note that a truly conventional pack without any string switches would save the 40 mΩ per-string MOSFET resistance; this baseline isolates the control policy, not the switch cost.

Scenario 2: Single-Cell Thermal Fault at 50% Mission

Purpose: Demonstrate fault isolation and graceful degradation.

Conditions: One cell in String A develops a thermal fault (temperature ramp at 5 °C/min, per Samsung SPEC 9.4) starting at 50% of mission. Samsung's discharge temperature limit is 75 °C; recommended derate release is 60 °C.

Metric	Fixed	Reconfigurable
Peak cell temperature	94.4 °C	60.0 °C
Isolation event	None	At 27.0 min
Mission energy	28.35 Wh	24.31 Wh
Runtime	40.8 min	33.9 min

Calculations:

- Fixed pack: No mechanism to disconnect the faulted cell. Temperature exceeded Samsung's stated maximum by 19.4 °C (94.4 – 75.0). The pack continued operating unsafely to deliver full energy.
- Reconfigurable pack: String A isolated when the faulted cell reached the 60 °C threshold. Mission continued on String B alone. Energy loss = 28.35 – 24.31 = 4.04 Wh (14.2%).

The 14% energy loss is the measured cost of safety. The fixed pack delivered more energy only because it ran a cell dangerously beyond its rated temperature with no protection.

Load continuity through the isolation event was confirmed — bus current dipped briefly and recovered within one control cycle (100 ms).

Scenario 3: 15% SOC Imbalance Between Strings

Purpose: Test whether reconfiguration extracts more usable energy from imbalanced strings.

Conditions: String A at 57.5% SOC, String B at 42.5% SOC (15% gap). Same 5 A load.

Metric	Fixed	Reconfigurable
Energy delivered	28.33 Wh	28.33 Wh
Join transient	—	8.00 A

Conclusion: No measurable advantage. Paralleled strings self-balance through the bus without any controller action. The SOC gap decayed to zero after joining. The milestone plan's hypothesis — that reconfiguration extracts measurably more usable energy from imbalanced strings — was tested and is not supported.

However, the reconfigurable topology did demonstrate safe connection of mismatched strings: the controller drained the fuller string first and joined at a controlled 1.5 A inrush, rather than an uncontrolled join that could produce up to 6.9 A.

Drive-Cycle Validation

Purpose: Confirm the state machine survives realistic accel/cruise/idle transients without false trips.

Conditions: 60 cycles of 15 A acceleration (0.5 s) → 5 A cruise → 0.5 A idle, total 2400 s.

Metric	Result
Total transitions	4 (normal startup sequence)
False overcurrent trips	0
Peak total current	15.00 A
Peak per-string	~7.5 A
Final fault	0

With both strings sharing the 15 A acceleration peaks, each string sees approximately 7.5 A — well below the 12 A per-string overcurrent threshold.

Scenario 4: Charging with SOC Imbalance — Theoretical Analysis

Purpose: Demonstrate independent charging paths as a structural advantage.

The Self-Balancing Problem

When two strings at different SOC's are connected in parallel for charging, the higher-SOC string discharges into the lower-SOC string through the inter-string current path before the

external charger's contribution matters.

For our 3S2P pack with String A at 50% SOC and String B at 98% SOC:

OCV at 50% ≈ 3.646 V/cell → string voltage = 10.938 V
OCV at 98% ≈ 4.122 V/cell → string voltage = 12.366 V
ΔV = 1.428 V
Total path resistance = R_stringA + R_stringB = 0.1 + 0.1 = 0.2 Ω
Inter-string current = 1.428 / 0.2 = 7.14 A

This 7.14 A flows from String B into String A uncontrolled, in addition to whatever the charger pushes. In our 2P pack, this happens to remain within the cell's 15 A rating, so the strings equalize without incident — which is why the simulation showed identical results for both topologies.

Why This Fails at Scale

The inter-string balancing current scales linearly with the number of high-SOC strings:

Configuration	Low string	High strings	ΔV	Current into low string
3S2P	1 × 50%	1 × 98%	1.43 V	7.1 A (2.4C)
3S4P	1 × 50%	3 × 98%	1.43 V	21.4 A (7.2C)
3S6P	1 × 50%	5 × 98%	1.43 V	35.7 A (12.1C)

At 6P, the weak string receives 35.7 A — over 12 times the cell's standard charge rate and 2.4 times its maximum continuous discharge rating. This would cause rapid heating, potential lithium plating, cell venting, and catastrophic failure. No fault condition is needed; the physics of self-balancing creates the hazard.

Reconfigurable Charging Eliminates This

With reconfigurable charging, strings are charged sequentially and independently:

1. Connect String A alone to the charger → charge at controlled 1C (3 A)
2. String A reaches 4.18 V/cell → disconnect A
3. Connect String B alone → charge at controlled 1C
4. String B reaches 4.18 V/cell → disconnect B → charge complete

Inter-string current: zero. Every cell charges at its safe rate regardless of the SOC gap.

Expected comparison at 50%/98% SOC imbalance:

Metric	Fixed Parallel	Reconfigurable Sequential
String A final SOC	~98.5% (limited by B hitting voltage ceiling)	100%
String B final SOC	~98.5%	100%
SOC spread	~0% (equalized through uncontrolled current)	0% (both independently full)
Max charge current	Uncontrolled (7.1 A in 2P, 35.7 A in 6P)	Charger-limited (3 A)
Lithium plating risk	High at large ΔV or high P-count	None — rate-controlled
Energy waste	I^2R loss in inter-string path	None — direct charger-to-string

3. Switching Subsystem Validation

Back-to-Back MOSFET Pair: Mandatory Design

The body diode of a single P-channel MOSFET provides a current path that bypasses the open switch. Only a back-to-back (common-source) pair blocks current in both directions.

Configuration	Leakage with switch open
Single IRF4905	168,571 mA
Back-to-back pair	0.0004 mA

Ratio: approximately 420 million to one. A single-MOSFET design cannot isolate a faulted string.

Precharge Requirement

Without precharge, closing a string switch onto a discharged bus capacitor (1000 μ F) produces an inrush of 218.8 A — 14.6 times the cell's continuous rating. A precharge resistor alone is insufficient when the load is connected:

Precharge configuration	Peak current
No precharge	218.8 A
R_pre = 1 Ω , load connected	36.4 A
R_pre = 10 Ω , load connected	89.9 A
R_pre = 10 Ω + load disconnect	1.09 A

Conclusion: A load disconnect switch is required hardware, not optional. This was discovered during simulation and added to the BOM.

4. Scaling Analysis: 2P vs 4P vs 6P

Fault Isolation — Advantage Grows with Scale

When one string is isolated, the fraction of capacity retained depends on the total number of parallel strings:

Configuration	Strings remaining	Capacity retained	Approx. energy after 50% fault
3S2P	1 of 2	50%	~75% of mission
3S4P	3 of 4	75%	~87.5%
3S6P	5 of 6	83%	~91.7%

The fixed pack story remains equally bad regardless of string count — it has no switch to isolate the fault.

Switch Resistance Overhead — Decreases with Scale

The per-string switch resistance (40 mΩ) is constant, but the per-string current drops as strings are added. Since loss scales with I², the total switch loss decreases:

For a 5 A total pack load:

Configuration	Current per string	Switch loss per string	Total switch loss
3S2P	2.50 A	0.250 W	0.500 W
3S4P	1.25 A	0.063 W	0.250 W
3S6P	0.83 A	0.028 W	0.167 W

Calculations:

- $P_{\text{switch}} = I^2 \times R_{\text{switch}} = I^2 \times 0.040 \, \Omega$
- 2P: $(2.5)^2 \times 0.040 \times 2 = 0.500 \, \text{W}$
- 4P: $(1.25)^2 \times 0.040 \times 4 = 0.250 \, \text{W}$
- 6P: $(0.833)^2 \times 0.040 \times 6 = 0.167 \, \text{W}$

Charging Safety — Critical at Scale

As shown in Section 2 (Scenario 4), inter-string self-balancing currents scale linearly with the number of high-SOC strings. At 6P with a 48% SOC gap, the weak string sees 35.7 A (12.1C) — a dangerous, uncontrolled condition. Reconfigurable sequential charging eliminates this entirely.

Wear Leveling — Only Possible at 3P and Above

With 2 strings, operation is binary: both or one. With 3+ strings, subset scheduling becomes possible — run 2 of 3 strings and rotate which one rests. This distributes cycle count across all strings and extends calendar life. The value is not more energy per cycle but longer total pack life.

5. Costs and Limitations — Stated Plainly

Hardware Costs of Reconfiguration

- $2 \times R_{DS(on)} = 40 \text{ m}\Omega$ per string, on top of $60 \text{ m}\Omega$ of cell resistance (67% increase in conduction path resistance)
- 6 MOSFETs (TO-220), 4 NPN level shifters, precharge resistor, bulk capacitor, load disconnect switch
- Additional ADC channels for per-string voltage and current sensing
- Firmware complexity: state machine with 13 states, debounced fault detection, precharge sequencing

Fundamental Limitation: Soft Degradation

A hard-paralleled pack masks a weak cell with its partner — the strong cell compensates and the pack continues with slightly reduced capacity. String-level isolation cannot do this: isolating one weak cell costs the two healthy cells in its string. This is an inherent tradeoff of string-level (vs cell-level) reconfiguration.

Known Gaps from Simulation

1. Inrush estimator ($\Delta V/R_{\text{string}}$) underpredicts by $\sim 5\times$ due to bus capacitor and load effects
 2. No thermal model — temperature is injected, not simulated; thermal claims are deferred to bench testing (M4)
 3. Coulomb counter drift: 5% seeded error grew to 5.06% after 10 min with no correction
 4. OCV-SOC table has $\sim 10\%$ discrepancy vs Samsung's shipping-voltage specification
 5. Capacity-fade lockout: if string capacities diverge, rejoin conditions may never be met simultaneously
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6. Conclusions

What Reconfiguration Provides (Measured)

1. **Physical fault isolation** — 0.0004 mA leakage with the back-to-back pair open, vs $168,571 \text{ mA}$ for a single switch. No internal current loop exists once the string switch opens.
2. **Graceful degradation** — fault at 27 min, isolated at 60°C (Samsung's recommended derate), mission continued on the healthy string. A fixed pack ran the cell to 94.4°C with no protection.

3. **Safe joining of mismatched strings** — precharge and controlled reconnection at bounded inrush (1.5 A measured). Generalises to different cell types, ages, and second-life cells.
4. **Independent charging paths** — eliminates uncontrolled inter-string currents that scale dangerously with parallel count. Enables safe charging of mismatched cells at controlled rates.

What Reconfiguration Does Not Provide

- No measurable energy extraction advantage from SOC imbalance (S3 disproved)
- No advantage under matched, fault-free conditions (S1 confirmed)
- Worse handling of soft cell degradation compared to hard-parallel topology

Framing

A hard-paralleled pack has no switch between cells, so a failing cell cannot be isolated and the pack fails as a unit. String-level reconfiguration inserts that switch — buying fault isolation, graceful degradation, safe joining of mismatched strings, and independent charging paths, at the cost of $2 \times R_{DS(on)}$ conduction loss and reduced tolerance to soft degradation. The 3S2P configuration represents the worst case for reconfiguration: isolating one of two strings sacrifices 50% of capacity. At higher parallel counts, the same isolation event retains 75–83% capacity, switch I^2R losses decrease quadratically, and subset scheduling enables wear-leveling strategies unavailable in a 2P topology. The safety and availability benefits demonstrated here therefore strengthen, not weaken, with scale.

Target Applications

String-level reconfiguration is not competitive with cell matching at automotive scale, where tight manufacturing tolerances and module-replacement strategies dominate. It addresses an unmet need in:

- **Second-life battery systems** — retired cells with varying capacity and health
 - **Remote/off-grid energy storage** — where technician access is limited and availability is critical
 - **Small EVs and e-mobility in emerging markets** — packs assembled from loose cells with unknown history
 - **Military/defense and medical equipment** — where mission completion and uptime outweigh hardware cost
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Appendix: Key Parameters

Parameter	Value	Source
Cell capacity	2.953 Ah	Verified, INTRO p.6
Cell R ₀	20 mΩ	Verified, INTRO p.3
Cell voltage range	2.50–4.20 V	Verified, INTRO p.3
MOSFET R _{DS(on)}	20 mΩ @ V _{GS} = -10 V	Verified, IRF4905 p.1-2
Body diode V _{SD}	1.3 V max	Verified, IRF4905 p.3
String resistance	100 mΩ	Calculated: 3×20 + 2×20
Precharge resistor	10 Ω	Design, validated Day 3
Bus capacitor	1000 μF	Assumption
OV trip	4.20 V/cell	Verified, SPEC 3.3
UV trip	2.50 V/cell	Verified, pack design guideline
OC trip	12 A/string	Design, below 15 A cell rating
OT trip	60 °C	Design, Samsung recommended derate
Charge-full threshold	4.18 V/cell	Design, 20 mV below OV trip
Precharge timeout	500 ms	Calculated, 10× time constant
Cooldown between changes	500 ms	Assumption
Debounce count	3 samples	Design